Arista Access Node Malleswaram SOP - ALARM_5978 - Routing Protocol Flap

Node: MALL-ACC-21 | **Vendor:** Arista | **Severity:** Major | **Alarm Code:** ALARM_5978

Description

The Routing Protocol Flap failure domain on the MALL-ACC-21 node is precipitated by a convergence of factors inherent to the Arista Access Node's implementation of the BGP, OSPF, and ISIS protocols. Specifically, the flapping of BGP routes, loss of OSPF adjacency, and ISIS LSP flooding are symptomatic of underlying issues such as routing table instability, protocol misconfiguration, or suboptimal hardware utilization. In this context, the Arista Access Node's reliance on the CLI for configuration and management necessitates a meticulous troubleshooting approach to identify and rectify the root cause of the failure domain.

Pre-requisite Checks

- Ensure remote hands are dispatched to the Malleswaram Node for physical access
- Verify the status of redundant links and ensure they are functioning nominally
- Check for any recent software or configuration updates that may have contributed to the failure

Potential Root Causes

Cause	Probability
Routing table instability due to protocol misconfiguration	High
Hardware utilization issues leading to protocol flapping	Medium
Software bugs or defects in the Arista Access Node firmware	Low

Diagnostic Commands & Expected Outputs

```
Command: show bgp summary, Expected Output: BGP router identifier: 192.168.1.1, Description: This command isolates the BGP routing table and identifies any potential issues with route flapping or instability.

Command: show ip ospf neighbor, Expected Output: Neighbor ID Pri State Dead Time Address Interface, Description: This command investigates OSPF adjacency loss and identifies any issues with neighbor relationships or link failures.

Command: show isis database, Expected Output: ISIS LSP Dump, Description: This command examines the ISIS LSP flooding and identifies any issues with link-state advertisement or database corruption.
```

Resolution Steps

Step 1: Step: Verify BGP routing table stability, Command: show bgp summary | grep -i 'flapping', Description: This command verifies that BGP route flapping has ceased and the routing table is stable.

Step 2: Step: Reset OSPF adjacency, Command: clear ip ospf neighbor, Description: This command resets the OSPF adjacency and re-establishes neighbor relationships.

Step 3: Step: Clear ISIS LSP database, Command: clear isis database, Description: This command clears the ISIS LSP database and re-establishes link-state advertisements.

Post-Check Verification

```
Command: show bgp summary | grep -i 'stable', Expected Output: BGP
router identifier: 192.168.1.1, Description: This command verifies
that the BGP routing table is stable and free of flapping routes.

Command: show ip ospf neighbor, Expected Output: Neighbor ID Pri State
Dead Time Address Interface, Description: This command verifies that
OSPF adjacency has been re-established and neighbor relationships are
stable.

Command: show isis database, Expected Output: ISIS LSP Dump,
Description: This command verifies that the ISIS LSP flooding has
ceased and the link-state advertisement database is stable.
```

Roll-back Procedures

- Step: Restore previous configuration, Command: copy running-config startup-config, Description: This command restores the previous configuration and reverts any changes made during the troubleshooting process.

- Step: Revert software or firmware updates, Command: show version | grep -i 'software', Description: This command reverts any recent software or firmware updates that may have contributed to the failure.